Progressive Refinement Network for Remote Sensing Image Change Detection

Xinghan Xu, Yi Liang, Jianwei Liu, Chengkun Zhang, and Deyi Wang

***Abstract*— Change detection (CD) in high-resolution remote sensing images (RSIs) aims at locating and understanding surface change areas. Despite some models have been proposed to solve the intrinsic problems of CD in RSIs (e.g., scale variation and internal nonconsistency), they resulted in less than ideal outcomes in specific scenes, such as objects with the same semantic concept but different spectrums and irrelevant change objects in the background. To this end, this article proposes a progressive refinement network (PRNet) to explore changes in more complex scenes in a continual calibration way. First, we excavate focused interactive deep semantic information with a proposed semantic refinement (SR) module based on the Vision Transformer and graph representation, which understands more useful semantic relations in ground objects. Second, we design a self-refinement (Self-R) module based on the supervised filtering framework to refine the shallow decoded features progressively. In addition, to ensure the structural information of the ground objects to the maximum extent, we propose a local detail enhancement (LDE) module based on multiscale convolutional architectures at the low-level encoding stage. Comprehensive experimental results on the two-instance RSI CD datasets and two public CD datasets demonstrate that the proposed PRNet achieves competitive performance with fewer parameters (3.44 M). Our code is available at https://github.com/lmetay/PRNet.**

***Index Terms*— Change detection (CD), filtering, graph representation, refinement, remote sensing images (RSIs), Transformer.**

I. Introduction

CHANGE detection (CD) allows understanding of the Earth's surface change from paired remote sensing images (RSIs), which are acquired with the same spatial domain at different temporal domains. The generated CD maps provide oriented surface knowledge for real-world projects, such as urban planning [1], sea reclamation [2], land-use detection [3], and so on.

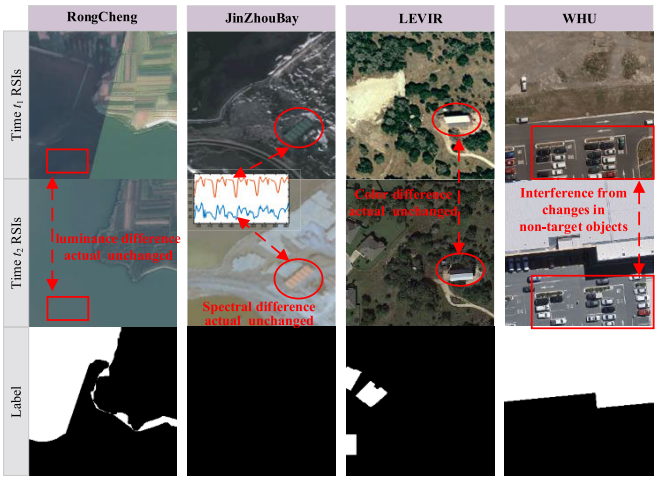


Fig. 1. Visualization of various CD scenes, in which RongCheng and JinZhouBay datasets focus on changes related to construction work along the coastal sea, while LEVIR and WHU datasets focus on building changes within the hinterland.

Recently, advanced optical sensors have collected more usable fine-scale production data, which provides rich spatial information. However, the ground objects with the same semantic interpretation in RSIs obtained at different spatial or temporal may exhibit significant spectral or fundamental feature differences. These differences arise from lighting, cloud cover, human activities, and other imaging environmental factors. As shown in the RongCheng sample of Fig. 1, the bottom-left sea area exhibits significant color differences due to illumination variations. Similarly, JinZhouBay and LEVIR samples show different spectra and colors on the roofs of the same construction due to human activities. In addition, high-solution RSI captures finely detailed with irrelevant changes, such as the clear vehicle details of the WHU sample. These problems pose great challenges for accurate CDs.

Currently, deep learning-based automatic interpretation technology for computer vision is increasingly emphasized in CD tasks [4]. Its main framework is to build the encoder–decoder (ED) based on either the convolutional neural network (CNN) [5] or the Transformer [6]. The encoder focuses more on general feature extraction of input RSIs, while the decoder emphasizes the discriminative feature learning about the change object. ED relies on multiple receptive fields and is effective for context modeling, with shallow features that provide detailed texture information and deep

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features that provide abstract semantic information. However, the general deep features from the encoder mainly represent content information of the objects rather than constructing change semantic links, resulting in insufficient cross-domain generalization. Based on this, many studies have been devoted to relational learning in terms of cross-domain feature fusion (CDFF) [7], [8] and semantic transfer (ST) [4], [9], which improves the performance of the model. Nevertheless, the CDFF complicates the computational efficiency, and related works have less discussion of the direct relationship between the feature scale and model performance. ST relies heavily on large-scale data and pretraining methods, which make the operation complex and affect model stability. Therefore, the object semantic relation modeling on pairwise deep features remains an open problem.

On the other hand, efficient filtering of irrelevant change is also one of the challenges of the CD task [10]. High-resolution RSI produces more and finer irrelevant changes, influenced by more advanced imaging sensors and imaging environments [11]. Affected by the high-altitude shooting and large-area coverage, the gaps of the irrelevant change objects in the scale and quantity make it harder to accurately identify pseudo-change areas [12]. Existing articles have developed multiple techniques (e.g., tensor decomposition, attention mechanism, knowledge guidance, etc.) to mitigate pseudo-changes in CD results. For example, Zhao et al. [13] achieved image content reason based on tensor CANDECOMP/PARAFAC decomposition theory to further filter pseudo-change. You et al. [14] proposed the horizontal feature fusion module based on the attention mechanism and multiscale depth-wise separable convolution to suppress pseudo-change. Lv et al. [15] introduced a change magnitude knowledge generated by sliding the adaptive region around the original input image pixel, which guides the feature extraction process to focus more on the change details. These techniques have performed satisfactorily in terms of accurately separating changed and unchanged areas, but they also sacrifice edge detail, and the modeling mechanism is not easily understood.

Based on the above analysis, this article proposes a progressive refinement network (PRNet) aimed at modeling deep semantic relationships and filtering irrelevant change objects to achieve CD more lightly and efficiently. To be specific, we propose a semantic refinement (SR) module based on the Vision Transformer [16] and graph representation to model the relationship between deep semantic features of pairwise RSIs and thus provide more generalized semantic information. Inspired by the success of the Transformer in capturing the spatial relationship [17], [18] and graph topology in flexibly representing spatial content [19], [20], we try to learn semantic relations through the Transformer and Vision Transformer. In this way, the model can efficiently refine deep semantic features and generate discriminative deep differential features at the high-level decoding stage. After that, we propose a self-refinement (Self-R) module based on the filtration framework to filter pseudo-changes and polish the details of change areas. The Self-R learns a mask to filter redundant information in the shallow differential features and continuously corrects this mask by a deep supervised strategy. In summary, our contributions are as follows.

1) We propose a PRNet for optical RSI change detection (CD) in a lightweight scheme. PRNet efficiently learns and progressively calibrates relevant change features based on the SR of deep image features and the Self-R of shallow difference features.
2) The SR module based on the Vision Transformer aims to model the semantic interaction relationships of paired images, thereby refining the deep image features from the CNN backbone based on the relationships. SR introduces graph node embedding to improve the higher-dimensional element-by-element representation of the Transformer, which characterizes the compact spatial information more expediently.
3) The Self-R module based on the supervised filtration framework aims to suppress the pseudo-change and sharpen the change areas. The guided Self-R learns a mask to update shallow difference features.

II. Related Works

A. CNN-Based CD Method

Compared to methods that rely only on surface information for discriminant analysis, such as arithmetic operations and feature transformation analysis, CNNs can characterize multiple levels of information about object content. The last decade has seen an explosion of countless CNN-based CD models, which are transferred from semantic segmentation and object detection tasks in model architecture, and the ED is a common framework [8], [21]. The familiar ED CNNs include fully convolution networks (e.g., FC-EF, FC-SiamConc, and FC-SiamDiff) [5], U-Net series (e.g., UNet and UNet++) [22], [23], high-resolution networks (HRNet) [24], [25], autonomously constructed networks based on ResNet [11], [26], or various convolutions (deconvolution, separable convolution, dilated convolution, grouped convolution, etc.) [27], [28], [29].

Pure CNNs learn the content information of an image incrementally through several fixed-size receptive fields. However, when dealing with some special scenes (such as narrow lines and large-scale areas in the RongCheng sample of Fig. 1), the features from pure CNNs are not sufficient to represent the objects due to the limited receptive fields. To this end, many methods establish connections between nonlocal pixels at the level of semantic content, instead of modeling neighbor pixel-relation in pure CNNs. These methods adopt technical including similarity metrics attention, semantic supervision strategy, and various Transformers. For example, Zhang et al. [30] embedded a cross-temporal difference attention module in the Transformer to perform semantic interaction by calculating the difference between the similarity matrix of cross-domain features and the similarity matrix of in-domain features. Zhang et al. [31] proposed dual-correlation attention in spatial and channel dimensions to build semantic relations between objects. Chen et al. [32] computed semantic loss based on the reconstructed feature of pairwise images to constrain semantic spaces in a self-supervised manner.

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Tang et al. [33] proposed a stacked Transformer to learn global semantic relationships. Li et al. [34] used a single Transformer to learn deep semantic relationships.

In addition to obtaining effective discriminative features, the discriminator in the model backend is also crucial as it directly affects the CD results [35]. Existing CD methods can be classified into metric-based CD (MCD) and classification-based CD (CCD) according to the discriminative mechanism. The MCD calculates the distance between the feature tensors of the RSIs (such as Euclidean distance [11], [26], Mahalanobis distance [36], [37], cosine distance [38], etc.) and combines with threshold separation to generate the CD results, which also leads to instability performance. The CCD obtains CD results by comparing the output probability maps of the fully connected layers or assembled convolutional layers. The CCD is more practical and reliable when supervised by labeling and is therefore widely used.

B. Transformer-Based CD Method

Vaswani et al. [39] proposed a Transformer based on the multihead self-attention (MSA) neural network for natural language processing (NLP) in 2017. Since the Transformer can effectively model long-range dependencies, it has recently been used in computer vision tasks, such as image segmentation [40], image super-resolution [41], image classification [42], object detection [43], and so on. However, Transformer involves the computation of a large number of self-attention scores (i.e., feature vector inner product), and thus requires a large computational space, especially for large feature sizes. Therefore, most studies in remote sensing combine CNNs and Transformers, rather than a purely Transformer-based model.

Existing papers of CD introduced Transformers to learn global contextual dependencies and to model relation change across temporal domains. For example, Chen et al. [44] proposed the bi-temporal image Transformer (BIT) to one-off capture global relationships of multiscale contextual features. Liu et al. [45] designed the token-aligned and semantic-aligned Transformers to achieve spatial and content alignment of RSIs with different resolutions. Zhang et al. [30] proposed cross-temporal difference attention instead of self-attention to learn relation change, which highlights the features of changed objects. Liu et al. [46] designed Transformers with cross-attention to calculate nonhomologous attention values and to achieve interactive learning of CNN features. Wang et al. [47] proposed a triplet Transformer framework to capture medium- and long-range dependencies from spatial attention, temporal attention, and spatial difference attention, respectively. Liu et al. [48] proposed a hierarchical Transformer structure from deep to shallow to fuse multiscale CNN features.

On the other hand, to improve computational efficiency, the initial Vision Transformer uses nonoverlapping and regular patch embedding as an input [16]. However, self-attention has a weak inductive bias for larger-size patch embedding and has redundant patch interactions [49]. Therefore, many studies have designed different embedding to suit their respective task needs. Chen et al. [44] designed token-vector embeddings similar to NLP based on spatial attention maps to represent visual semantics. Song et al. [50] designed a progressive sampling semantic token module to iteratively generate new semantic token embeddings to alleviate the structural damage caused by the original patch splits. Xu et al. [51] proposed spatial attention-based visual tokenizers to obtain semantic-level embeddings.

III. Proposed Model

A. Architecture Overview

Fig. 2 illustrates the PRNet architecture that includes a Siamese encoder for five-scale feature extraction and a progressive feature refinement decoder for differential feature learning. The procedure of PRNet can be illustrated by Algorithm 1.

First, the input RSIs ($\{\mathbf{X}^{(t_1)}, \mathbf{X}^{(t_2)}\} \in \mathbb{R}^{3\times H\times W}$) are fed into the Siamese encoder to gradually extract the half-size encoded features $\{\mathbf{F}^{*}_{e,k}\}^{5}_{k=1} \in \mathbb{R}^{64\times H_k\times W_k}$ ($* \in \{t_1, t_2\}$), where k is the feature scale index, the channel dimensions are all 64, $H_{k+1} = H_k/2$, and $W_{k+1} = W_k/2$ (the details as illustrated in the encoders of Fig. 2 and Step1 of Algorithm 1). Second, $\mathbf{F}^{*}_{e,4}$ and $\mathbf{F}^{*}_{e,5}$ are sent to the two SR modules, respectively. SR is used to perform two RSIs deep semantic information interaction and modeling and output the SR features ($\mathbf{F}^{*}_{\mathrm{SR},4}$ and $\mathbf{F}^{*}_{\mathrm{SR},5}$) (the details illustrated in Fig. 3 and described in Step 2 of Algorithm 1). Two deeper decoder convolutional (Dconv) blocks fuse SR features and generate differential features ($\mathbf{F}_{d,5}$ and $\mathbf{F}_{d,4}$). In particular, the deepest Dconv uses an all-zero tensor of the same size as $\mathbf{F}^{*}_{e,5}$ as input instead of $\mathbf{F}_{d,i+1}$. Then, the three shallow Dconvs progressively generate shallow differential features ($\{\mathbf{F}_{d,k}\}^{3}_{k=1}$), which revised from Self-R modules to $\{\mathbf{F}_{\mathrm{Self}-R,k}\}^{3}_{k=1}$. Self-R learns masks in a supervised manner to suppress irrelevant changes (the details as illustrated in Fig. 4 and Step3 of Algorithm 1). Finally, a simple convolutional-based classifier (Conv in Fig. 2 consists of a 3×3 convolutional layer with Batch Normalization (BN) and ReLU activation, a 1×1 convolutional layer, a sigmoid layer, and a $2\times$upsampling layer) is used to output the change probability map $\hat{\mathbf{Y}}_p \in \mathbb{R}^{2\times H\times W}$, and the predicted CD map $\hat{Y} \in \mathbb{R}^{H\times W}$ is obtained based on a pixel-wise Argmax on the channel dimension of $\hat{\mathbf{Y}}_p$ (the details as illustrated in Step4 of Algorithm 1).

In addition, we extract more useful detail information by expanding the local receptive fields with the local detail enhancement (LDE) module in the low-level encode stage. LDE is formed by multiscale convolutional layers with BN and ReLU (i.e., 3×3, 5×5, and 7×7 kernels) and a 1×1 convolutional layer with BN used for downsampling. To transfer the output features ($\{\mathbf{F}_{\mathrm{lde},k}\}^{2}_{k=1} \in \mathbb{R}^{64\times H_k\times W_k}$) of the LDE to the subsequent encoding stage, a simple companion device (concatenation operation $+$ 1×1 convolutional layer with 64 output channels) was employed. During training, the predicted CD map and label are used to compute the change loss, while the output of the mask by the three Self-R modules and label are used to compute the Self-R loss.

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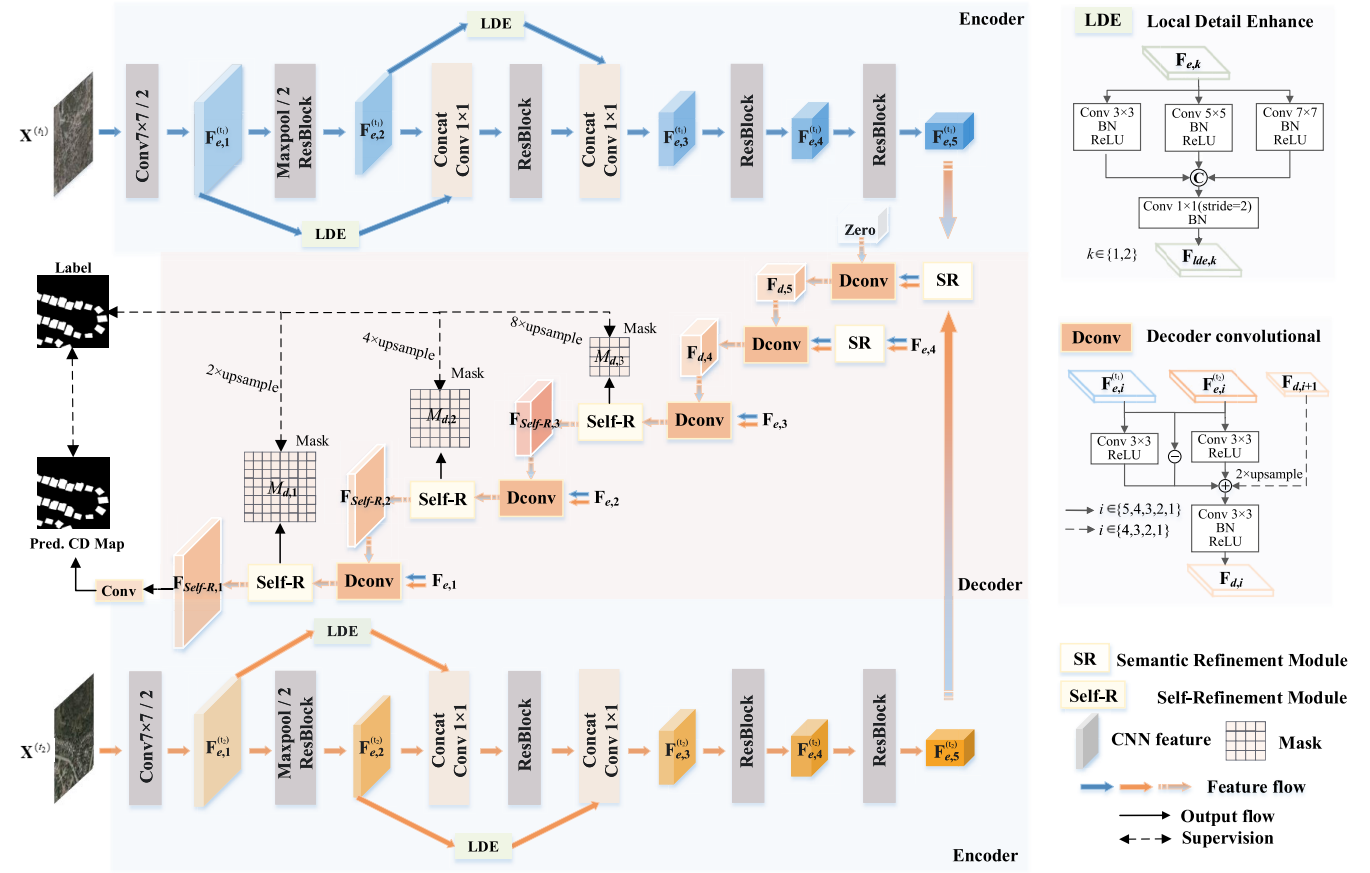


Fig. 2. Architecture of the proposed PRNet.

B. SR Module

To achieve a more compact semantic interaction and efficient internal relationship modeling, we introduce the SR module based on Transformer and graph representation. The presentation of SR was inspired by BIT proposed by Chen et al. [44]. The difference is that we use graph representation instead of the token vectors of spatial attention simplified CNN features in BIT. In this way, graph representation both establishes long-range spatial connections and simplifies spatial relationships no matter what kind of object is, making it more suitable for CD tasks with macrosemantic concepts. As shown in Fig. 3, the purpose of SR is to revise deeper CNN features $\mathbf{F}_{e,k}^{(t_1)}$ and $\mathbf{F}_{e,k}^{(t_2)}$ ($k = 4$ or 5) to refined features $\mathbf{F}_{\mathrm{SR},k}^{(t_1)}$ and $\mathbf{F}_{\mathrm{SR},k}^{(t_2)}$. The SR process consists of three important steps: 1) graph-domain projection; 2) semantic relation learning; and 3) refinement and domain inverse projection, whose details are as follows.

1) Graph-domain projection aims to convert CNN feature $\mathbf{F}_{e,k}^{*} \in \mathbb{R}^{C\times H_k\times W_k}$ into graph-structured data $G_{e,k}^{*} \in \mathbb{R}^{N\times L}$, where C is the channels of the CNN feature, N is the number of nodes of the graph data, and L is a node dimension [52]. In other words, CNN features with the size of $C\times H_k\times W_k$ are compressed into N nodes, and each node is represented by a vector with the size of $L\times 1$. We use a 2-D 1×1 convolutional layer and a 1-D convolutional layer (the output channel are N and L, respectively) to perform graph-domain projection in spatial and channel dimensions, expressed as follows:

$$G_{s,k}^{*} = \phi\left(\mathbf{F}_{e,k}^{*}\right) = W_\phi \times \mathbf{F}_{e,k}^{*} \tag{1}$$

$$G_{c,k}^{*} = \varphi\left(\mathbf{F}_{e,k}^{*}\right) = W_\varphi \times \mathbf{F}_{e,k}^{*} \tag{2}$$

where $G_{s,k}^{*}$ is the graph representation from a spatial perspective, then flattened into a 2-D tensor $G_{s,k}^{*} \in \mathbb{R}^{N\times H_k W_k}$. We also flatten and reshape $\mathbf{F}_{e,k}^{*}$ to a 2-D tensor, which is sent to 1-D convolutional layer to obtain the graph representation from channel perspective, that is, $G_{c,k}^{*} \in \mathbb{R}^{L\times H_k W_k}$. $\phi(\cdot)$ denotes 1×1 convolution function. $\varphi(\cdot)$ denotes a 1-D convolution function with a kernel size of 1. $W_\phi \in \mathbb{R}^{N\times C}$ and $W_\varphi \in \mathbb{R}^{L\times C}$ denote the weights of two convolutional layers. Then, $G_{e,k}^{*} = G_{s,k}^{*} \times (G_{c,k}^{*})^{\mathrm{T}}$ indicates the graph state of the feature $\mathbf{F}_{e,k}^{*}$.

2) Semantic relation learning aims to achieve relational learning between pairwise RSIs at the semantic level. Transformer is used to learn the relationships of graph nodes $G_{e,k}^{(t_1)}$ and $G_{e,k}^{(t_2)}$ to characterize the semantic relation. We concatenate $G_{e,k}^{(t_1)}$ and $G_{e,k}^{(t_2)}$ into a graph set $G_k \in \mathbb{R}^{2N\times L}$ and add the learnable graph-positional embedding $P_k \in \mathbb{R}^{2N\times L}$ to G_k as an input (i.e., $G_k' = G_k + P_k$) to Transformer (based on Vision Transformer). The Transformer consists of two stages, each of which contains two important components, that is, a prior layer normalization (LN) and a posterior MSA, and a prior LN and a posterior multilayer perceptron (MLP). Skip connections are adopted to integrate the different state information of the nodes.

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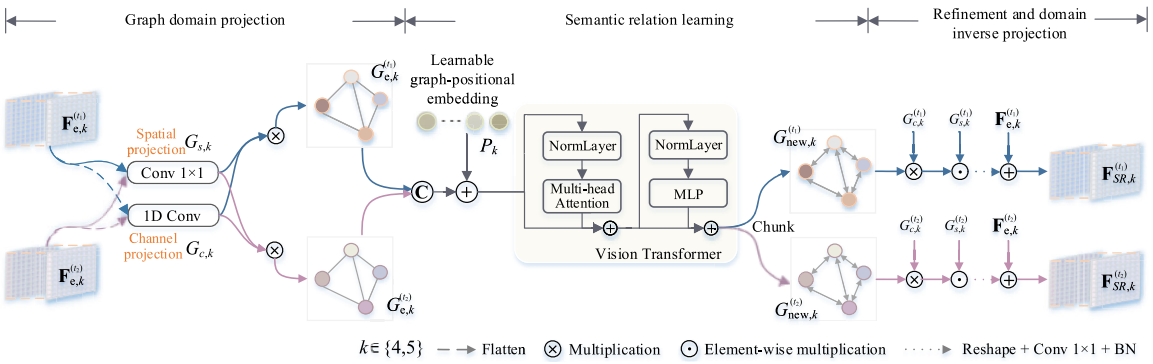


Fig. 3. Structure of the SR module.

Algorithm 1 Implementation Process of PRNet

Input: Input RSIs: $\mathbf{X}^{(t_1)}, \mathbf{X}^{(t_2)}$;
Output: Predicted change map: $\hat{Y}$;

```
// Step1 : Extraction of Dual − temporal image features
for t in {t1, t2} do
    F_{e,1}^{(t)} = Conv_{7×7}(X^{(t)})
    for k in {1, 2, 3, 4, 5} do
        if k == 1 then
            F_{lde,k}^{(t)} = LDE(F_{e,k}^{(t)})
            F'_{e,2}^{(t)} = Conv_{1×1}(Cat(ResBlock(F_{e,1}^{(t)}, F_{lde,k}^{(t)}))
        end if
        if k == 2 then
            F_{lde,k}^{(t)} = LDE(F_{e,k}^{(t)})
            F_{e,3}^{(t)} = Conv_{1×1}(Cat(ResBlock(F'_{e,2}^{(t)}), F_{lde,k}^{(t)}))
        end if
        if k ≥ 4 then
            F_{e,k}^{(t)} = ResBlock(F_{e,k−1}^{(t)})
        end if
    end for
end for
// Step2 : Semantic refinement
for k in {5, 4} do
    F_{SR,k}^{(t1)}, F_{SR,k}^{(t2)} = SR(F_{e,k}^{(t1)}, F_{e,k}^{(t2)})
end for
F_{d,5} = Dconv(F_{SR,5}^{(t1)}, F_{SR,5}^{(t2)}, 0)
F_{d,4} = Dconv(F_{SR,4}^{(t1)}, F_{SR,4}^{(t2)}, F_{d,5})
// Step3 : Self − refinement
for k in {3, 2, 1} do
    if k == 3 then
        F_{Self−R,k} = Self−R(Dconv(F_{e,k}^{(t1)}, F_{e,k}^{(t2)}, F_{d,4}))
    else
        F_{Self−R,k} = Self−R(Dconv(F_{e,k}^{(t1)}, F_{e,k}^{(t2)}, F_{Self−R,k+1}))
    end if
end for
// Step4 : Outputting the predicted change map
Ŷ_p = Conv(F_{Self−R,1})
Ŷ = Argmax(Ŷ_p)
```

MSA calculates and weights the dot product scores with multiple self-attention mechanisms to measure important relationships between nodes. Given the nodes set G'_k that have been layer normalized, three linear transformations were used to output the different states of G'_k. That is, query $\mathbf{Q}_k$, key $\mathbf{K}_k$, and value $\mathbf{V}_k$. The linear projection is represented as follows:

$$\mathbf{Q}_k, \mathbf{K}_k, \mathbf{V}_k = G'_k \times {W_k}^{Q}, G'_k \times {W_k}^{K}, G'_k \times {W_k}^{V} \tag{3}$$

then, reshaping query, key, and value to 3-D tensors, that is, $\mathbf{Q}_k \in \mathbb{R}^{N_h \times 2N \times D_h}$, $\mathbf{K}_k \in \mathbb{R}^{N_h \times 2N \times D_h}$, and $\mathbf{V}_k \in \mathbb{R}^{N_h \times 2N \times D_h}$, where N_h is the number of heads and is set to 8 and D_h is the dimension of $\mathbf{Q}_k$, $\mathbf{K}_k$, and $\mathbf{V}_k$ and is set to 64. ${W_k}^{Q}$, ${W_k}^{K}$, ${W_k}^{V} \in \mathbb{R}^{L \times (N_h * D_h)}$ are the weights of the three linear layers. The MSA score is calculated as follows:

$$\mathbf{S}_k = \mathrm{SA}(\mathbf{Q}_k, \mathbf{K}_k, \mathbf{V}_k) = \mathrm{Softmax}\left(\frac{\mathbf{Q}_k \mathbf{K}_k^{\mathrm{T}}}{\sqrt{D_h}}\right)\mathbf{V}_k. \tag{4}$$

In contrast to general self-attention, MSA jointly characterizes the full range of subspace information by performing multiple independent self-attention and concatenating all the results. We use a linear transformation to project the score $\mathbf{S}_k \in \mathbb{R}^{N_h \times 2N \times D_h}$ back into the graph representation again, and residual connect it to the input graph G'_k

$$G_{\mathrm{up},k} = \mathbf{S}_k \times W_\psi + G'_k \tag{5}$$

where $W_\psi \in \mathbb{R}^{(N_h * D_h) \times L}$ is the weight of the linear layer. $G_{\mathrm{up},k} \in \mathbb{R}^{2N \times L}$ is graph state after MSA. Next, $G_{\mathrm{up},k}$ is sent to the second stage (an LN and an MLP) of the Transformer to further encode the node relationships. MLP includes two linear layers and a Gaussian error linear unit (GeLU) activation. The learning process is as follows:

$$\begin{aligned} G_{\mathrm{new},k} &= \mathrm{MLP}\left(G_{\mathrm{up},k}\right) + G_{\mathrm{up},k} \\ &= \mathrm{GeLU}\left(G_{\mathrm{up},k} \times W_\mu\right) \times W_\nu + G_{\mathrm{up},k} \end{aligned} \tag{6}$$

where $G_{\mathrm{new},k} \in \mathbb{R}^{2N \times L}$ is the set of the node states output by the Transformer. GeLU$(\cdot)$ denotes GeLU activation. $W_\mu \in \mathbb{R}^{L \times 2N}$ and $W_\nu \in \mathbb{R}^{2N \times L}$ are the weights of the two linear layers in the MLP, respectively. Finally, $G_{\mathrm{new},k}$ is split into two chunks, that is, $G_{\mathrm{new},k}^{(t_1)}, G_{\mathrm{new},k}^{(t_2)} \in \mathbb{R}^{N \times L}$.

3) Refinement and domain inverse projection aim to use Transformer-based semantic relations to refine the input graph and convert back graph-structured data into CNN features. we use multiplication to refine the initial channel graph $G^*_{c,k}$,

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and element-wise multiplication to refine the initial spatial graph $G^*_{s,k}$

$$G^*_{\mathrm{SR},k} = G^*_{\mathrm{new},k} \times G^*_{c,k} \odot G^*_{s,k} \tag{7}$$

where $G^*_{\mathrm{SR},k} \in \mathbb{R}^{N \times H_k W_k}$ is a graph refined for semantic relation. $\odot$ is element-wise multiplication broadcasted along feature planes. Finally, we use a 1×1 convolutional layer with BN for domain inverse projection and skip connection to bridge the input and refined CNN features

$$\mathbf{F}^*_{\mathrm{SR},k} = \mathrm{BN}\left(\vartheta\left(G^*_{\mathrm{SR},k}\right)\right) + \mathbf{F}^*_{e,k} = \mathrm{BN}\left(W_\vartheta \times G^*_{\mathrm{SR},k}\right) + \mathbf{F}^*_{e,k} \tag{8}$$

where $\mathbf{F}^*_{\mathrm{SR},k} \in \mathbb{R}^{C \times H_k \times W_k}$ is grid feature refined for semantic relation. $\vartheta(\cdot)$ and $W_\vartheta \in \mathbb{R}^{C \times N}$ are the convolution function and weight for domain inverse mapping, respectively. BN$(\cdot)$ is batch normalization.

C. Self-R Module

Since the encoder focuses more on the representation of the general feature of the input RSIs without explicit change attributes, thus less identified redundancy. Accordingly, the decoder will gradually present redundancy and the pseudo-change. To suppress the phenomenon, we design the Self-R module to achieve shallow differential feature cleaning with supervised modulation. Self-R adopts a filtration framework to generate a mask to further update the differential features fused by the decoder convolutional block. As shown in Fig. 4, shallow differential features $\mathbf{F}_{d,k} \in \mathbb{R}^{64 \times H_k \times W_k}$ $(k = \{1, 2, 3\})$ is fed to a 3×3 convolutional block with BN and ReLU activation (no change in output channel) to encode it. Then, a 1×1 convolutional layer (the output channel is 32), a 3×3 convolutional layer (the output channel is 1), and a sigmoid function to obtain the mask $M_{d,k} \in \mathbb{R}^{H_k \times W_k}$ for multiplication. The above process is represented by

$$M_{d,k} = \delta\left(W_m \times \mathbf{F}_{d,k}\right) \tag{9}$$

where $W_m \in \mathbb{R}^{1 \times 32}$ denotes the weights product of three convolutional layers. $\delta(\cdot)$ is a sigmoid function. Finally, $M_{d,k}$ encodes the changing attribute of $\mathbf{F}_{d,k}$, and Self-R adopts skip connection and outputs the refined features $\mathbf{F}_{\mathrm{Self}-R,k} \in \mathbb{R}^{64 \times H_k \times W_k}$

$$\mathbf{F}_{\mathrm{Self}-R,k} = \mathbf{F}_{d,k} \odot M_{d,k} + \mathbf{F}_{d,k}. \tag{10}$$

To achieve effective Self-R, the mask $M_{d,k}$ is constrained by the label during the network training stage, rather than creating $M_{d,k}$ out of nothing.

D. Loss Function

The total loss L includes two components, the binary CD loss L_c and the Self-R loss $L_{s,k}$ $(k = \{1, 2, 3\})$. L_c represents the the difference between the predicted CD map $\hat{Y}$ and the label Y. $L_{s,k}$ represents the the difference between the kth-scale mask $M_{d,k}$ and Y. In particular, the bilinear interpolation is employed to upsample 2^k times for $M_{d,k}$ to match the size of Y. We use a combination of dice loss and binary cross-entropy

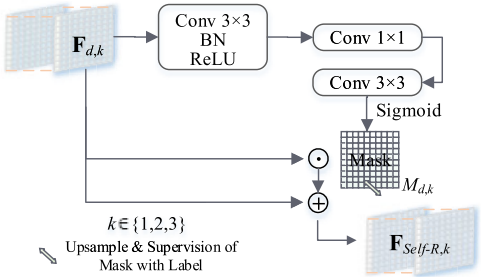


Fig. 4. Structure of the Self-R module.

loss to calculate the L_c and $L_{s,k}$ on each pixel, which are calculated as follows:

$$L = L_c + \sum_{k=1}^{3} L_{s,k} = f\left(\hat{Y}, Y\right) + \sum_{k=1}^{3} f\left(M_{d,k}, Y\right) \tag{11}$$

$$f\left(\hat{x}, x\right) = f_{\mathrm{bce}}\left(\hat{x}, x\right) + f_{\mathrm{dice}}\left(\hat{x}, x\right) \tag{12}$$

$$f_{\mathrm{bce}}\left(\hat{x}, x\right) = \sum_{i=1,j=1}^{H \times W} \frac{x_{ij}\log\hat{x}_{ij} + \left(1 - x_{ij}\right)\log\left(1 - \hat{x}_{ij}\right)}{H \times W} \tag{13}$$

$$f_{\mathrm{dice}}\left(\hat{x}, x\right) = 1 - \sum_{i=1,j=1}^{H \times W} \frac{2\hat{x}_{ij}x_{ij} + \varepsilon}{\hat{x}_{ij} + x_{ij} + \varepsilon} \tag{14}$$

where $\hat{x}$ and x are denoted as the predicted value and the label, respectively. i and j are the position indices of the pixels. ε is a smoothing constant that prevents the denominator from being 0 and is set to 0.0001. $x_{ij} = 1$ $(x_{ij} = 0)$ indicates the pixel that is changed (unchanged) pixel for the label.

IV. Experiments

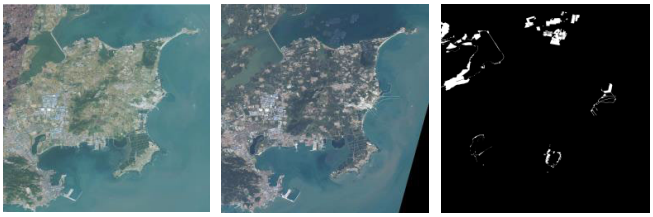
A. Datasets and Implementation Details

We conduct comprehensive experiments on two instance datasets and two public datasets, including: **1)** RongCheng-CD; **2)** JinZhouBay-CD; **3)** LEVIR-CD [26]; and **4)** WHU-CD [53], which are described as follows.

1) RongCheng-CD: The dataset contains a pair of nonhomologous RSIs recording the changes in seaside farms in RongCheng, China. RSIs were collected from the Gaofen-2 satellite in October 2017 and the Gaofen-1 satellite in September 2022, which have a resolution of 1 and 2 m/pixel, respectively. We downsample time t_1 RSI to 2 m/pixel to match time t_2 RSI, as shown in Fig. 5. The matched RSIs with the size of 11554×11554 are cropped to 256×256 patches with an overlap on the right and bottom. Some invalid patches (the black part without content in Time t_2 RSI) are removed, and the remaining patches are randomly divided into 1353 pairs for training, 193 pairs for validation, and 390 pairs for testing.

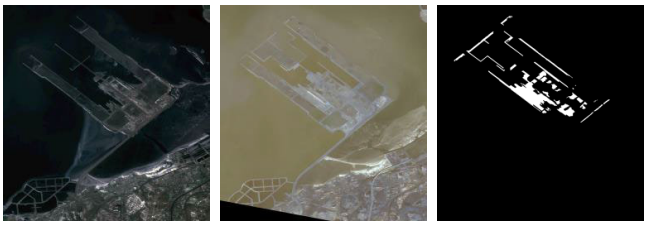
2) JinZhouBay-CD: The dataset contains a pair of nonhomologous RSIs recording the land reclamation in Jinzhou-Bay, China. RSIs were collected from the Gaofen-1 satellite in February 2015 and the Gaofen-2 satellite in January 2017, which have a resolution of 2 and 1 m/pixel. We downsample time t_2 RSI to 2 m/pixel to match time t_1 RSI, as shown in Fig. 6. The matched RSIs with the size of 5667×5423 are

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(a) (b) (c)

Fig. 5. RongCheng-CD dataset. (a) Time t_1 RSI of RongCheng collected in 2017. (b) Time t_2 RSI of RongCheng collected in 2022. (c) Label.



(a) (b) (c)

Fig. 6. JinZhouBay-CD dataset. (a) Time t_1 RSI of JinzhouBay collected in 2015. (b) Time t_2 RSI of JinzhouBay collected in 2017. (c) Label.

cropped to 256 × 256 patches with an overlap on the right and bottom and randomly divided these patches into 311 pairs for training, 44 pairs for validation, and 90 pairs for testing after removing invalid patches.

3) LEVIR-CD: The dataset contains 637 pairs of RSIs covering various buildings in the city, with a size of 1024 × 1024 and a resolution of 0.5 m/pixel. We use the training set, validation set, and test set provided by the original authors in a 7:1:2 ratio and crop the RSIs with the size of 256 × 256.

4) WHU-CD: The dataset contains one pair of RSIs recording urban building changes before and after the earthquake disaster, with size of 32508 × 15354 and a resolution of 0.3 m/pixel. We crop the original paired image into 256 × 256 patches with an overlap on the right and bottom and randomly divide these patches into 5460 pairs for training, 779 pairs for validation, and 1561 pairs for testing.

We perform all comprehensive experiments by using PyTorch on a PC with an Intel Core i9-13900KF 3.00-GHz CPU, 24-GB RAM, and an NVIDIA RTX 3090Ti GPU. In the training stage, the parameters of PRNet are randomly initialized by Kaiming initialization. The Adam algorithm is used to optimize PRNet with a batch size of 16, and the initial learning rate is 0.001 in the first 50 epochs and decays linearly to e^{-7} in the follow 50 epochs. In the graph domain projection of two SRs, the number of graph nodes N are set to be 2^2 and 4^2 (4^2 and 4^2) corresponding to the SRs with $\mathbf{F}^*_{e,4}$ and $\mathbf{F}^*_{e,5}$ as inputs in RongCheng and JinZhouBay (LEVIR and WHU) datasets, which are discussed in Section V-A. The node dimension used to characterize the node content is set to $L = 2 \times N$.

In addition, we use the following metrics, including overall accuracy (OA), precision (Pre), recall (Rec), $F1$-measure ($F1$), Kappa coefficient (KC), and mean intersection-over-union (mIoU) to quantitatively evaluate the performance of the model. Validation is transacted after training in each epoch, and then the best model on the validation set evaluates the performance of the model on the test set.

TABLE I

QUANTITATIVE COMPARISON STUDIES WITH SOTA CD MODELS ON THE RONGCHENG DATASET [**RED BOLD** HIGHLIGHTS THE **BEST** SCORE, **BLUE BOLD** HIGHLIGHTS THE **SECOND-RANKED** SCORE, AND ALL RESULTS ARE EXPRESSED AS THE PERCENTAGE (%)]

Models	RongCheng					
	OA	Pre	Rec	F1	KC	mIoU
FC-EF	98.30	65.08	63.55	64.30	63.43	72.83
FC-SiamConc	98.58	68.58	75.69	71.96	71.23	77.38
DSAMNet	98.44	63.31	**83.82**	72.15	71.35	77.41
BIT	99.02	85.78	70.96	77.67	77.17	81.25
P2V	**99.13**	**90.06**	71.73	**79.85**	**79.41**	**82.79**
MSCANet	98.64	75.69	64.18	69.46	68.77	75.92
DMINet	99.02	**85.95**	70.75	77.61	77.12	81.21
USSFCNet	98.95	81.66	72.69	76.92	76.38	80.71
ADHRNet	98.83	83.83	63.44	72.23	71.64	77.67
ICIFNet	98.69	74.48	69.54	71.92	71.25	77.41
A2Net	99.00	84.52	71.55	77.49	76.98	81.12
PRNet (Ours)	**99.05**	81.47	78.51	**79.96**	**79.48**	**82.82**

B. Comparison With State-of-the-Art Methods

We compare the proposed PRNet with 11 state-of-the-art (SOTA) CD models, including FC-EF [5], FC-SiamConc [5], DSAMNet [11], BIT [44], P2V [54], MSCANet [48], DMINet [55], USSFCNet [56], ADHRNet [24], ICIFNet [57], and A2Net [58]. To be fair, we reproduce these comparison methods using their default hyperparameters and the same experimental setup as our study. The quantitative and qualitative comparative results are analyzed as follows.

1) Comparison on the RongCheng Dataset: Table I reports the all quantitative comparison results on the RongCheng test sets. From it, PRNet scores highest in $F1$/KC/mIoU (i.e., 79.96%/79.48%/82.82%), only 0.08% lower the best P2V in OA, and scores more intermediate in Pre and Rec. More significantly, PRNet slightly outperforms the second-ranked P2V. This shows that both are effective in learning the changed features of the seaside farm. However, PRNet's equivalent performance in Pre and Rec is more stable than P2V with higher Pre and lower Rec, indicating that PRNet has strong robustness to the imbalance of changed to unchanged pixels.

Fig. 7 shows all qualitative comparison results. It can be seen that PRNet has the fewest false detections (red and blue) and renders cleaner visuals. For scenes with narrower change areas, such as the **1st**, **2nd**, and **4th** rows, PRNet can accurately locate the change areas and achieves relatively fine-grained detection. For scenes heavily influenced by the imaging environment, such as the insignificant foreground and background in $\mathbf{X}^{(t_1)}$ of **3rd** row, and the shadowed large areas in $\mathbf{X}^{(t_2)}$ of **5th** row, the CD results of PRNet prove its ability to better prevent the influences and did not exhibit widespread false positives.

2) Comparison on the JinZhouBay Dataset: From Table II, PRNet achieves the best performance in OA/Rec/$F1$/KC/mIoU (i.e., 99.44%/89.80%/92.46%/92.17%/92.70%). The $F1$/KC/mIoU of PRNet is 6.20%/6.41%/5.28% higher than that of the second-ranked USSFCNet. The gain gap of Rec between PRNet and second-ranked P2V is more significant, that is,

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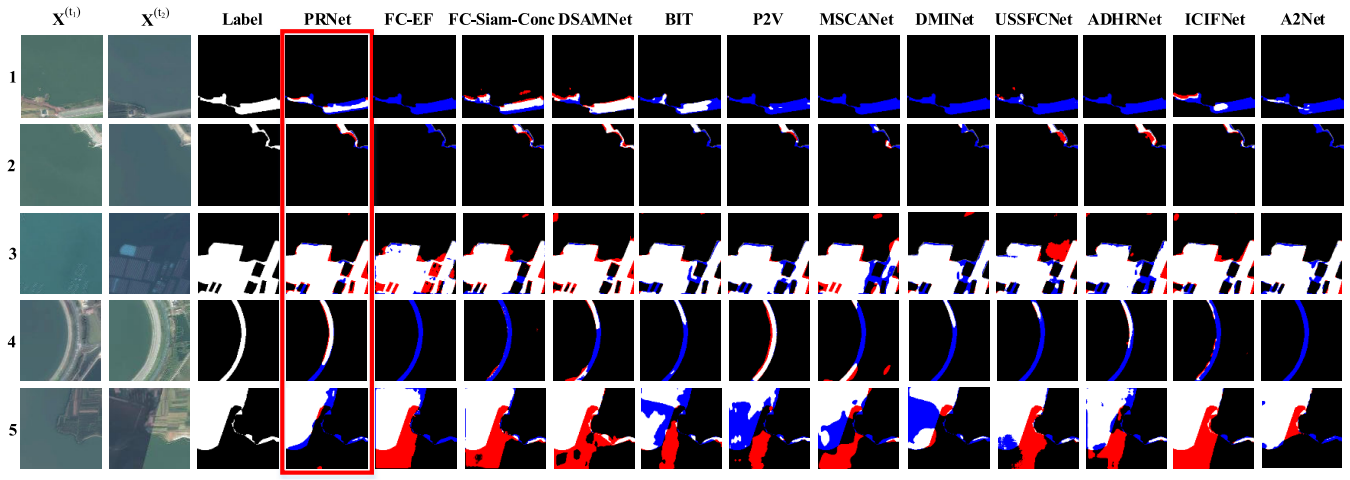


Fig. 7. Visual comparison of PRNet and SOTA models on the RongCheng test set. Where white represents true positive, black represents true negative, red represents false positive, and blue represents false negative.

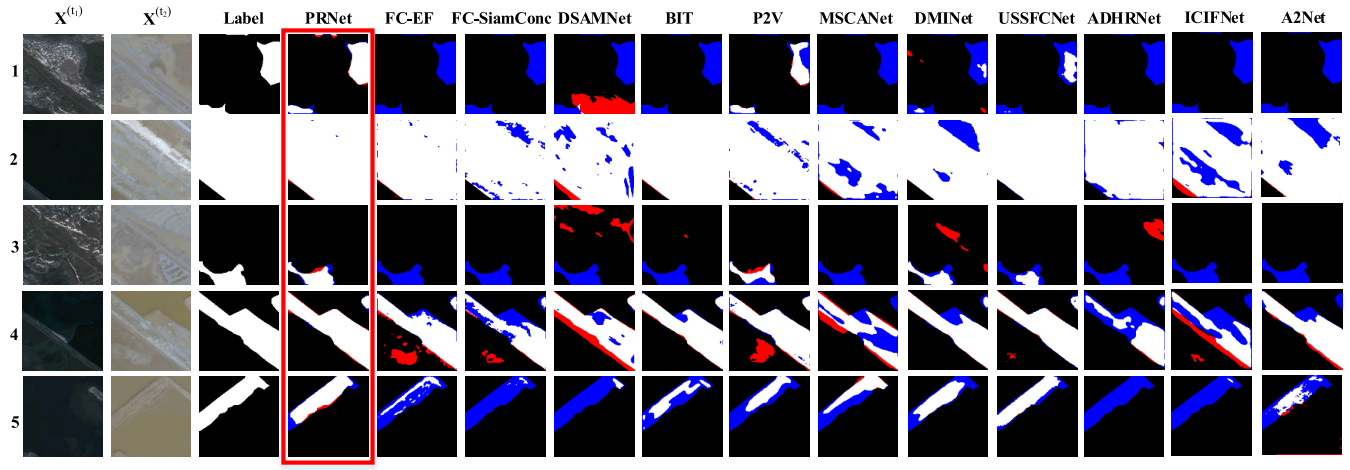


Fig. 8. Visual comparison of PRNet and SOTA models on the JinZhouBay test set.

10.14% higher. This demonstrates that PRNet's ability to identify changed pixels far exceeds that of other models.

From Fig. 8, it can be seen that PRNet outperforms comparison models in scenes with large-scale changed areas. Furthermore, land reclamation is a complex project that involves abundant construction materials. Therefore, many scenes do not show significant change features (such as the **1st** $\sim$ **3rd** rows), which makes detection more difficult and results in many comparison models not locating change areas. PRNet, on the other hand, not only locates the change areas but also obtains a finer boundary close to the label. This may be because based on the enhanced discriminant features via semantic interaction and relationship modeling, PRNet pays more attention to the concept of change in the SR stage.

3) Comparison on the LEVIR Dataset: From Table III, PRNet performs the second best in OA/Rec/$F1$/KC/mIoU, but has little difference in overall performance compared with the best ADHRNet. Specifically, the $F1$/KC/mIoU scores lose the percentage gain of 0.19%/0.21%/0.18% compared with the ADHRNet. The reason is that the difference in Rec between the two is insufficient to offset the disparity in Pre between them. However, ADHRNet needs to consume a lot of computational resources according to Model Efficiency Analysis in Section V-C. In addition, it is obvious that PRNet obtains improved results than other relation-learning CD models (BIT, MSCANet, USSFCNet, and ICIFNet).

TABLE II

QUANTITATIVE COMPARISON STUDIES WITH SOTA CD MODELS ON THE JINZHOUBAY DATASET [**RED BOLD** HIGHLIGHTS THE **BEST** SCORE, **BLUE BOLD** HIGHLIGHTS THE **SECOND-RANKED** SCORE, AND ALL RESULTS ARE EXPRESSED AS THE PERCENTAGE (%)]

Models	JinZhouBay					
	OA	Pre	Rec	F1	KC	mIoU
FC-EF	98.07	76.81	71.38	73.99	72.99	78.37
FC-SiamConc	97.96	83.30	58.81	68.95	67.92	75.26
DSAMNet	97.03	61.70	60.49	61.09	59.55	70.47
BIT	98.80	95.21	72.49	82.31	81.70	84.35
P2V	98.95	91.86	**79.66**	85.33	84.78	86.66
MSCANet	97.06	63.09	56.72	59.73	58.21	69.79
DMINet	98.39	92.39	63.25	75.09	74.29	79.23
USSFCNet	**99.03**	94.26	79.51	**86.26**	**85.76**	**87.42**
ADHRNet	97.93	82.62	58.62	68.58	67.54	75.04
ICIFNet	96.56	56.91	43.44	49.27	47.52	64.59
A2Net	97.48	75.68	51.04	60.97	59.72	70.64
PRNet (Ours)	**99.44**	**95.29**	**89.80**	**92.46**	**92.17**	**92.70**

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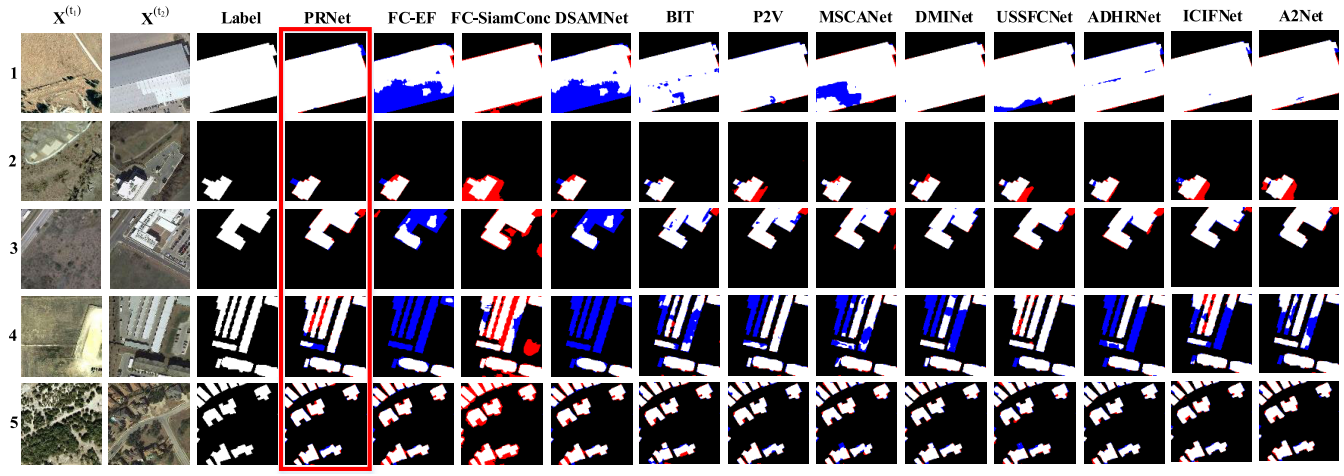


Fig. 9. Visual comparison of PRNet and SOTA models on the LEVIR test set.

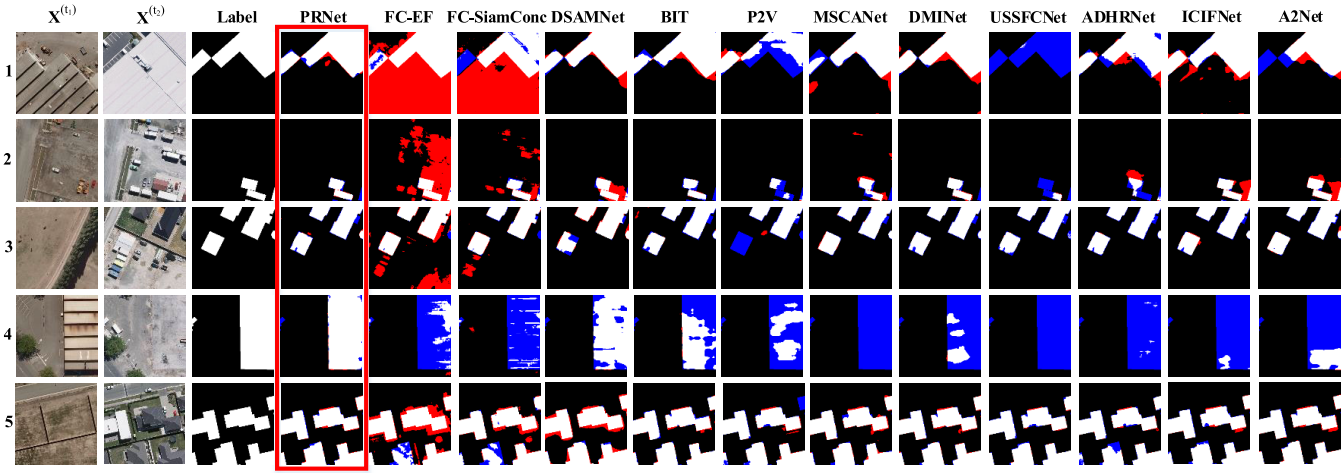


Fig. 10. Visual comparison of PRNet and SOTA models on the WHU test set.

TABLE III

QUANTITATIVE COMPARISON STUDIES WITH SOTA CD MODELS ON THE LEVIR DATASET [**RED BOLD** HIGHLIGHTS THE **BEST** SCORE, **BLUE BOLD** HIGHLIGHTS THE **SECOND-RANKED** SCORE, AND ALL RESULTS ARE EXPRESSED AS THE PERCENTAGE (%)]

Models	LEVIR					
	OA	Pre	Rec	F1	KC	mIoU
FC-EF	96.32	61.09	76.68	68.00	66.08	73.85
FC-SiamConc	96.59	60.19	**97.35**	74.39	72.67	77.81
DSAMNet	98.18	85.62	77.12	81.15	80.19	83.19
BIT	98.93	92.12	86.30	89.11	88.55	89.62
P2V	98.97	92.57	86.64	89.51	88.96	89.96
MSCANet	98.81	88.58	88.01	88.30	87.67	88.90
DMINet	99.00	92.18	87.89	89.98	89.46	90.37
USSFCNet	98.98	**93.25**	86.29	89.63	89.10	90.08
ADHRNet	**99.05**	**93.42**	87.47	**90.35**	**89.85**	**90.70**
ICIFNet	98.98	91.89	87.72	89.76	89.22	90.18
A2Net	99.00	91.77	88.18	89.94	89.41	90.33
PRNet (Ours)	**99.01**	91.18	**89.16**	**90.16**	**89.64**	**90.52**

Fig. 9 shows that PRNet can better avoid false detections. In contrast, PRNet not only exhibits fewer omissions of shadow-obscured changing buildings but also ensures the complete overall structure of the objects. Specifically, DMINet, USSFCNet, ICIFNet, and A2Net each demonstrate the best visual results in a number of examples, such as DMINet, ICIFNet, and A2Net obtain almost the same CD maps as labels in the **1st** row, USSFCNet obtains the better performance in the **3rd** and **4th** rows, and A2Net also generates the complete change area in the **3rd** row. While PRNet performs more consistently and portrays finer edges due to Self-R with supervised modulations in the shallow decode stage. For the scene shown in the **4th** row, the CD map of PRNet produces minor sticking, which is due to the dense building roofs having very similar features to the ground. In short, PRNet also has strong capabilities in internal relationship modeling for large-scale objects.

4) Comparison on the WHU Dataset: It can be seen from Table IV that PRNet obtains a 0.45%/0.45%/0.40% gain in $F1$/KC/mIoU compared to the second-placed USSFCNet. USSFCNet obtains the highest Pre (96.16%) but the difference of PRNet between Pre and Rec is less than that of USSFCNet. This indicates that USSFCNet oversuppresses the background and fails to detect more changed pixels. Subsequently, among FC-EF/SiamConc, DSAMNet, MSCANet, and ICIFNet have lower Pre and higher Rec, which indicates they are not strong enough to suppress background pixels.

Fig. 10 also shows the superior performance of PRNet in relation to modeling and suppression of pseudo-change.

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TABLE IV

QUANTITATIVE COMPARISON STUDIES WITH SOTA CD MODELS ON THE WHU DATASET [**RED BOLD** HIGHLIGHTS THE **BEST** SCORE, **BLUE BOLD** HIGHLIGHTS THE **SECOND-RANKED** SCORE, AND ALL RESULTS ARE EXPRESSED AS THE PERCENTAGE (%)]

Models	WHU					
	OA	Pre	Rec	F1	KC	mIoU
FC-EF	96.91	52.41	**93.40**	67.14	65.65	73.67
FC-SiamConc	97.93	64.06	87.96	74.13	73.08	78.38
DSAMNet	98.33	68.05	**95.42**	79.45	78.60	82.09
BIT	99.33	88.77	91.60	90.16	89.81	90.70
P2V	99.47	**94.83**	89.19	91.93	91.65	92.26
MSCANet	98.74	76.78	90.03	82.88	82.23	84.73
DMINet	99.37	89.27	92.37	90.78	90.45	91.23
USSFCNet	**99.51**	**96.16**	88.98	**92.43**	**92.18**	**92.71**
ADHRNet	99.22	87.17	90.16	88.64	88.24	89.40
ICIFNet	98.97	81.18	90.52	85.59	85.06	86.88
A2Net	99.48	91.94	92.75	92.34	92.07	92.62
PRNet (Ours)	**99.53**	94.68	91.14	**92.88**	**92.63**	**93.11**

The most obvious example, the **2nd** row, shows multiple infrastructures that can be easily confused with the lower right change building, whereas PRNet eliminates the interference and accurately identifies the changing areas. For the scene with multiple changed buildings, such as the **5th** row, PRNet is capable of detecting buildings of various shapes and positions without missing any. In summary, PRNet has obvious advantages in associating multiple objects, identifying large-scale objects, and suppressing background interference.

C. Ablation Study

To verify the role of different network components of PRNet and different loss functions, we conduct a series of ablation studies. The results are analyzed as follows.

1) Validation of Different Network Components: This part of the ablation studies used the same training configuration as PRNet. The network components of PRNet include the SR module, the Self-R module, and the LDE module, resulting in three simplified models, which are described as follows.

1) Removing two LDEs and their companion device from the encoder, which is denoted as **w/o LDEs**.
2) Removing two SRs acting on the encoded features $\mathbf{F}^*_{e,4}$ and $\mathbf{F}^*_{e,5}$ ($* \in \{t_1, t_2\}$), which denoted as **w/o SRs**.
3) Removing three Self-R acting on the shallow differential features $\{\mathbf{F}_{d,k}\}_{k=1}^{3}$, which denoted as **w/o Self-Rs**.

From Table V, each component contributes significantly to the model performance. Specifically, comparing the results of w/o LDEs and PRNet proves that LDEs bring the relative gains of $F1$, KC, and mIoU and are about 0.19∼3.06%, 0.20∼3.16%, and 0.16∼2.68% in four datasets. SRs bring the relative gains of $F1$, KC, and mIoU and are about 0.95∼6.26%, 0.99∼6.49%, and 0.86∼5.34%. Self-Rs bring the relative gains of $F1$, KC, and mIoU and are about 1.39∼3.88%, 1.46∼4.04%, and 1.20 ∼3.40%. The contribution of SR is significantly better than that of LDE and Self-R on the RongCheng and JinZhouBay datasets. This is because SR bridges the excellent semantic relationship within the change concept in the case that blurred image content is affected by aquatic background and construction materials. On the other hand, Self-R contributes significantly better than

TABLE V

ABLATION STUDIES ABOUT DIFFERENT COMPONENTS IN PRNET

Models	Params(M)	RongCheng			JinZhouBay		
		F1	KC	mIoU	F1	KC	mIoU
w/o LDEs	1.97	78.52	78.04	81.83	89.40	89.01	90.02
w/o SRs	3.35	76.34	75.72	80.25	86.21	85.68	87.36
w/o Self-Rs	3.37	77.99	77.47	81.44	88.58	88.13	89.30
PRNet	3.44	79.96	79.48	82.82	92.46	92.17	92.70
Models	**FLOPs(G)**	**LEVIR**			**WHU**		
		F1	KC	mIoU	F1	KC	mIoU
w/o LDEs	5.18	89.97	89.44	90.36	92.26	92.00	92.55
w/o SRs	19.75	89.20	88.63	89.69	91.93	91.64	92.25
w/o Self-Rs	19.33	88.77	88.18	89.32	91.45	91.17	91.84
PRNet	19.76	90.16	89.64	90.52	92.88	92.63	93.11

LDE and SR on the LEVIR and WHU datasets. This is because Self-R ensures a finer and completely changed building and suppresses pseudo-changes.

Fig. 11 shows the qualitative comparison results of the simplified models. It can be seen that the w/o LDEs model shows incomplete detection for large-scale change areas due to the inability to efficiently learn local contextual information during the feature extraction. The w/o SRs model shows a large miss detection (blue) due to the lack of ability to learn a system of semantic relations. The w/o Self-Rs model shows more false detections near the boundary of change areas (red), especially in scenes with significant pseudo-changes (such as the **2nd** row of the RongCheng dataset, and the **1st** row of the JinZhouBay dataset). In contrast, PRNet obtains CD maps with internal consistency and fewer false detections, demonstrating that the combination of three network components is beneficial for spatial features modeling and suppression background.

To measure the computational cost of the different components, we calculated model parameters (Params) and floating-point operations per second (FLOPs) (based on RSI with the size of $3 \times 256 \times 256$) for three simplified models and listed in Table V. The results show PRNet has 3.44 M params and 19.76 G FLOPs, where SR and Self-R require very low computational space and LDE increases the computational burden. Considering the important role of the multiscale convolutional structure of LDE in learning large change areas, we sacrifice some acceptable computational costs.

2) Validation of Different Loss Functions: This section verifies the constraining role of the Self-R loss $L_{s,k}$ ($k = \{1, 2, 3\}$) on mask $M_{d,k}$. Since PRNet needs to learn 3 scales $M_{d,k}$, we construct 4 loss function combinations: 1) only the binary CD loss L_c; 2) $L_c + L_{s,3}$; 3) $L_c + L_{s,2} + L_{s,3}$; and 4) $L_c + L_{s,1} + L_{s,2} + L_{s,3}$ [the total loss as described by formula (11)]. The quantitative comparison results on the four datasets are shown in Table VI. It shows that PRNet trained by total loss achieves the best performance except for the LEVIR dataset. In the LEVIR dataset, the gradual introduction of $L_{s,k}$ is not ideal for improving the detection accuracy of PRNet, which may be because some details of the dense small-scale change areas are interfered with by the upsampling of $M_{d,k}$. However, the total loss trained PRNet in LEVIR has a small gap to the best scores, that is, $F1$/KC/mIoU are 0.11%/0.11%/0.10% lower than $L_c + L_{s,2} + L_{s,3}$, which

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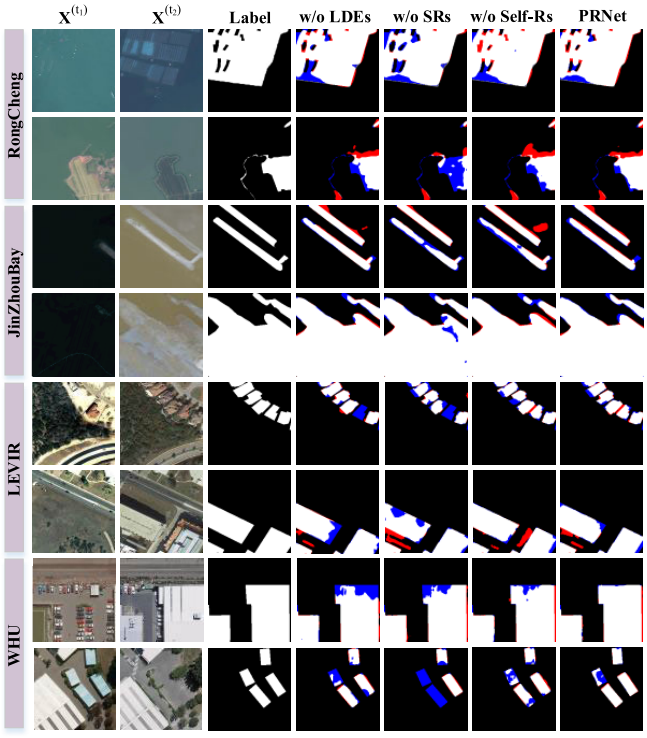


Fig. 11. Visual comparison of the different components of PRNet.

TABLE VI
ABLATION STUDIES ABOUT DIFFERENT LOSS FUNCTIONS IN THE TRAINING STAGE

Loss functions	RongCheng			JinZhouBay		
	F1	KC	mIoU	F1	KC	mIoU
L_c	78.57	78.05	81.83	87.35	86.85	88.27
L_c+$L_{s,3}$	79.53	79.05	82.53	90.29	89.92	90.78
L_c+$L_{s,2}$+$L_{s,3}$	79.82	79.35	82.74	91.04	90.70	91.44
L_c+$L_{s,1}$+$L_{s,2}$+$L_{s,3}$	**79.96**	**79.48**	**82.82**	**92.46**	**92.17**	**92.70**

Loss functions	LEVIR			WHU		
	F1	KC	mIoU	F1	KC	mIoU
L_c	90.08	89.55	90.45	91.82	91.54	92.16
L_c+$L_{s,3}$	90.01	89.48	90.39	92.35	92.08	92.63
L_c+$L_{s,2}$+$L_{s,3}$	**90.27**	**89.75**	**90.62**	92.27	92.01	92.56
L_c+$L_{s,1}$+$L_{s,2}$+$L_{s,3}$	90.16	89.64	90.52	**92.88**	**92.63**	**93.11**

is allowable. In summary, imposing the constraint strategy on $M_{d,k}$ is effective.

V. DISCUSSION

A. Sensitivity Analysis of the Hyperparameter

The graph node of the SR spatially tokens the dense pixel-level features into a compact graph topology. In this section, we conducted a series of crossover experiments to explore the influence of the number of graph nodes N on PRNet. We use N_4 and N_5 to denote the number of nodes in the SRs corresponding to $F^*_{e,4} \in \mathbb{R}^{64\times16\times16}$ and $F^*_{e,5} \in \mathbb{R}^{64\times8\times8}$ in this section. Considering the sizes of $\mathbf{F}^*_{e,4}$ and $\mathbf{F}^*_{e,5}$, we set the range of the N_4 to $\{2^2, 4^2, 8^2\}$ and the range of the N_5 to $\{2^2, 4^2\}$, where the squares indicate horizontal and vertical dimensions with no real physical meaning. Fig. 12 shows the $F1$ score of PRNet under different N_4 and N_5. In the RongCheng dataset, the $F1$ scores of PRNet show a monotonic change as N_4 increases when fixing N_5. In the JinZhouBay dataset, the $F1$ scores gradually decrease as N_4 increases and this decay is faster for larger values of N_5, indicating that the more finely tagged $\mathbf{F}^*_{e,5}$ interferes more with the CD results for simple image content. In the LEVIR and WHU datasets, $F1$ scores take on increase first and then decrease with little magnitude as N_4 increases, proving a sparse or dense graph topology will also hinder the model performance. In conclusion, a compact graph topology is useful for bridging semantic relations of interest changes. Based on the above analysis, we set $N_4 = 2^2$ and $N_5 = 4^2$ on the RongCheng and JinZhouBay datasets, and $N_4 = 4^2$ and $N_5 = 4^2$ on the LEVIR and WHU datasets.

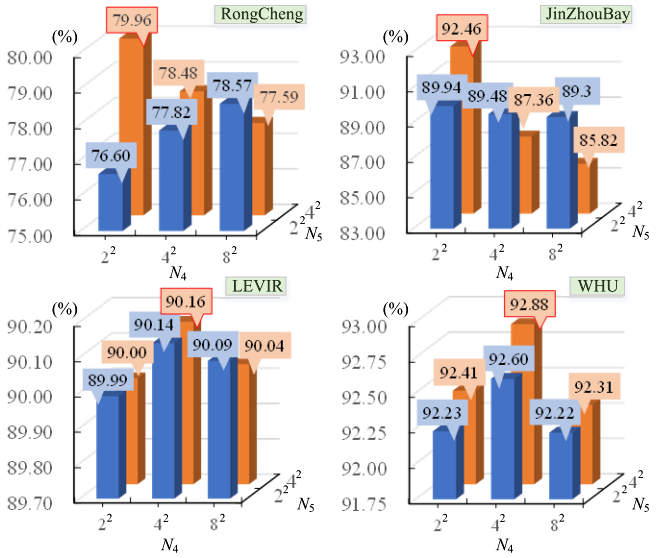


Fig. 12. $F1$ scores of PRNet for different numbers of graph nodes in SR.

B. Model Intermediate Feature Visualization

To better comprehend our model, we use the Grad-Cam method to visualize the feature activation maps at each period of PRNet and reshape them as 256×256. Taking a sample from the LEVIR test set as an example, the feature activation maps of PRNet are shown in Fig. 13. Given the input RSIs $\mathbf{X}^{(t_1)}$ and $\mathbf{X}^{(t_2)}$, the Siamese encoder extract the five-scales encoded features $\{\mathbf{F}^*_{e,k}\}^5_{k=1}$ $(* \in \{t_1, t_2\})$. Where low-level encoded features $\{\mathbf{F}^*_{e,k}\}^3_{k=1}$ capture visual characteristics such as structure of the salient objects, high-level encoded features $\{\mathbf{F}^*_{e,k}\}^5_{k=4}$ provide semantic information such as ground object categories. Then, SRs perform semantic interaction relationship modeling, refine the high-level encoded features, and output SR features $\{\mathbf{F}^*_{\mathrm{SR},k}\}^5_{k=4}$. Compared with $\{\mathbf{F}^*_{e,k}\}^5_{k=4}$, $\{\mathbf{F}^*_{\mathrm{SR},k}\}^5_{k=4}$ can highlight the areas of semantic concepts related to the change or unchanged and enhance more compact space representation. After that, the decoder gradually generates difference features $\{\mathbf{F}_{d,k}\}^5_{k=1}$, and Self-Rs effectively filter or blur irrelevant change areas in low-level difference features $\{\mathbf{F}_{d,k}\}^3_{k=1}$. Finally, the change probability map $\hat{\mathbf{Y}}_p$ accurately locates the change areas.

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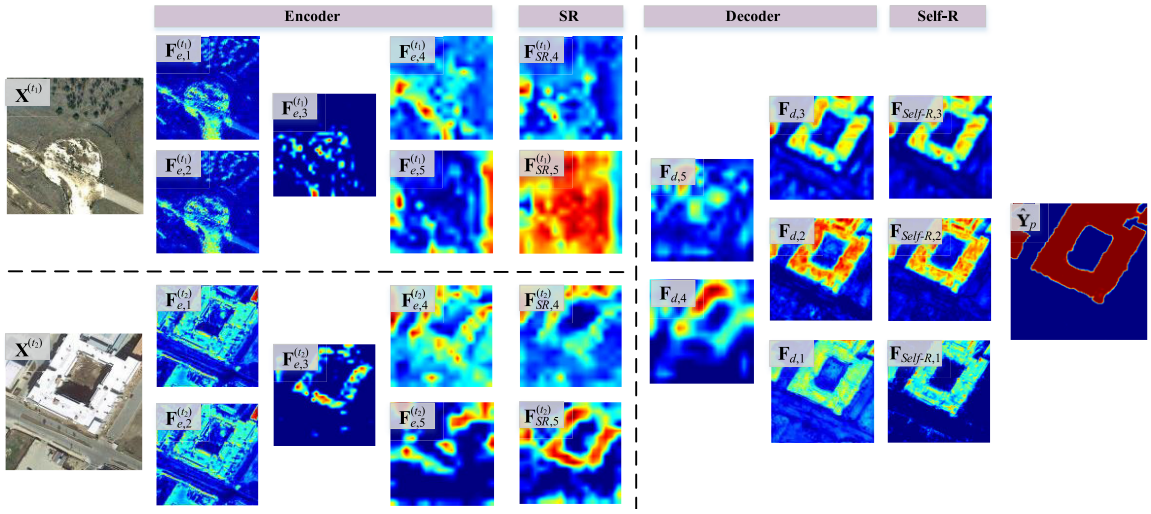


Fig. 13. Visualization feature maps of the PRNet.

TABLE VII
COMPARISON OF COMPUTATIONAL COST OF DIFFERENT MODELS

Models	Params(M)	FLOPs(G)	Time(ms)
FC-EF	1.35	3.57	4.63
FC-SiamConc	1.55	5.32	5.75
DSAMNet	16.95	75.35	11.18
BIT	3.01	8.45	5.90
P2V	5.42	32.86	7.81
MSCANet	16.59	14.70	12.43
DMINet	6.76	14.19	9.32
USSFCNet	1.52	4.84	20.96
ADHRNet	12.90	119.48	25.17
ICIFNet	25.83	25.28	36.33
A2Net	3.78	6.02	12.50
PRNet	3.44	19.76	11.17

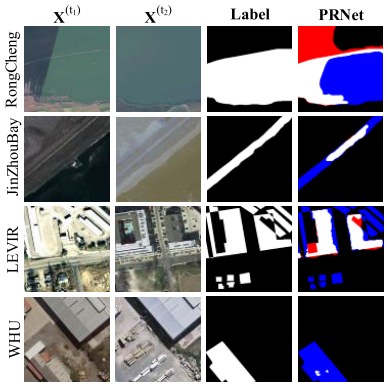


Fig. 14. Visualization of the failure cases. (a) Time t_1 RSIs. (b) Time t_2 RSIs. (c) Label. (d) CD maps generated by PRNet.

C. Model Efficiency Analysis

Table VII lists the efficiency of PRNet and SOTA models in terms of Params, FLOPs, and inference time (Time), where Params and FLOPs are calculated as described in the ablation study, and Time refers to the time consumed by the forward propagation of a pair of RSIs. The results show that PRNet has lower parameters (3.44 M), centered FLOPs (19.76 G), and Time (11.17 ms). Combined with the results in Table V, it can be inferred that PRNet has higher FLOPs and Time due to multiscales convolutional parallelized implementation in LDE. Nevertheless, we have not removed LDE due to it improves the $F1$ scores by 0.19%∼3.06% on four datasets. Overall, PRNet enjoys more accurate CD results with the sacrifice of a few computational costs.

D. Analysis of Failure Cases

Although PRNet shows a continually progressing performance, there are still unsatisfactory CD maps for some challenging scenes. Fig. 14 shows some representative failure cases for the following reasons: 1) the presence of imperceptible change areas and 2) the blends in with the background. As shown in the examples of the RongCheng and JinZhouBay datasets, submerging of aquaculture rafts and filling of construction did not exhibit significant visual changes, resulting in large-range missed alarms. In the examples of the LEVIR and WHU datasets, the changed building roof and a changed area made up of scattered simple buildings show similar colors to the public ground, resulting in many missed detections. In addition, a deeper common reason for the failure cases is also the limited number of similar samples to those in Fig. 14, which is insufficient and allows us to improve the feature extraction capability of our model. To this end, the introduction of knowledge migration or camouflaged object detection techniques under the premise of ensuring computational cost as far as possible can be worked as future research.

VI. CONCLUSION

In this article, we devote to performing CD in high-resolution optical RSIs and propose the PRNet, which can efficiently construct semantic relations of interest change for image content and suppress the irrelevant changes. We propose the Transformer-based SR module to learn the advanced graph-relation representations to refine deep features with semantic concepts of change interest, which improves the recognition of change objects. Moreover, we propose a Self-R module based on the supervised filtration framework to polish

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the details of change areas and filter irrelevant changes. Experimental evaluations over two instance datasets and two public datasets indicate that PRNet has excellent performance, higher compatibility, and fewer parameters (3.44 M). Nevertheless, some insufficient limiting samples and low contrast between change objects and backgrounds hinder the accuracy of PRNet in dealing with more complex scenes. In the future, it is worth exploring the knowledge transfer or the camouflage object detection techniques in CD tasks to optimize the model.

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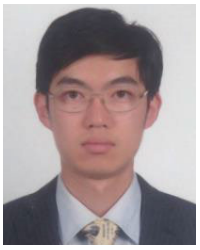
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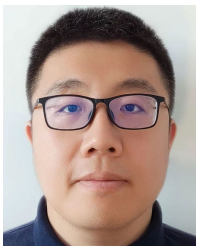
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